

SIMATS ENGINEERING
CSA1503-CLOUD COMPUTING

22.To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.

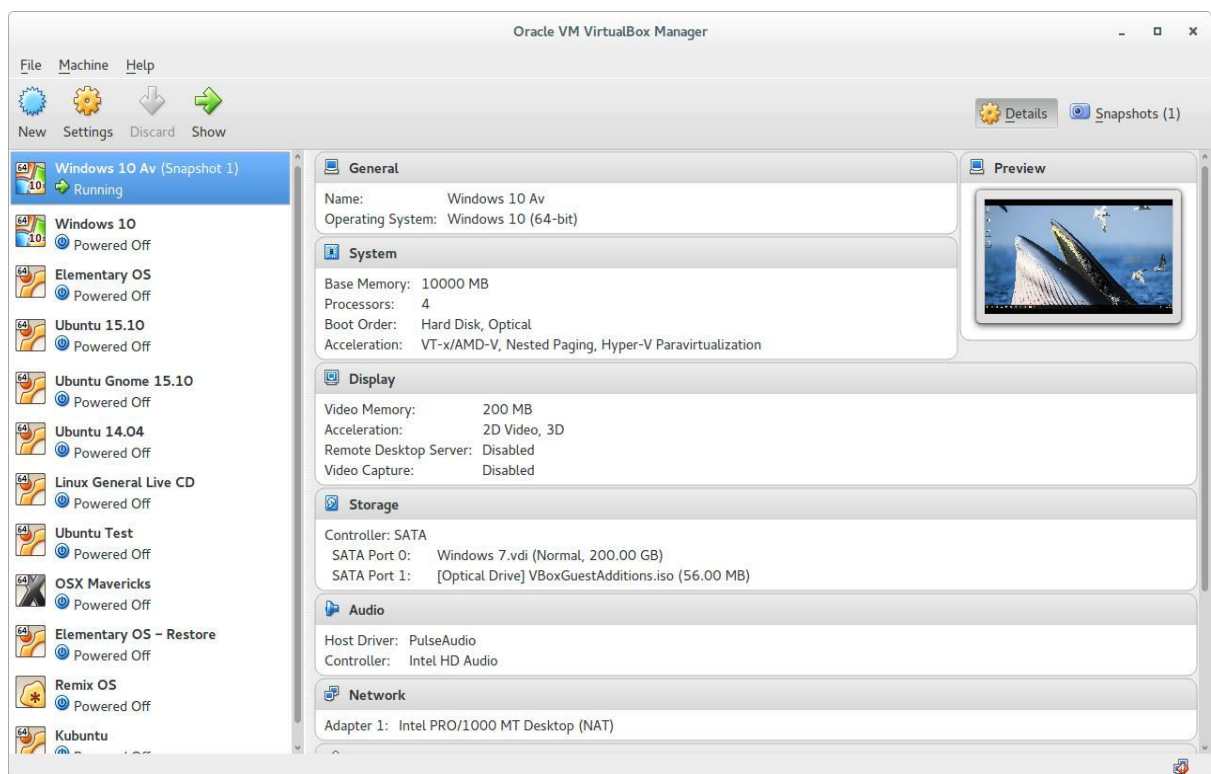
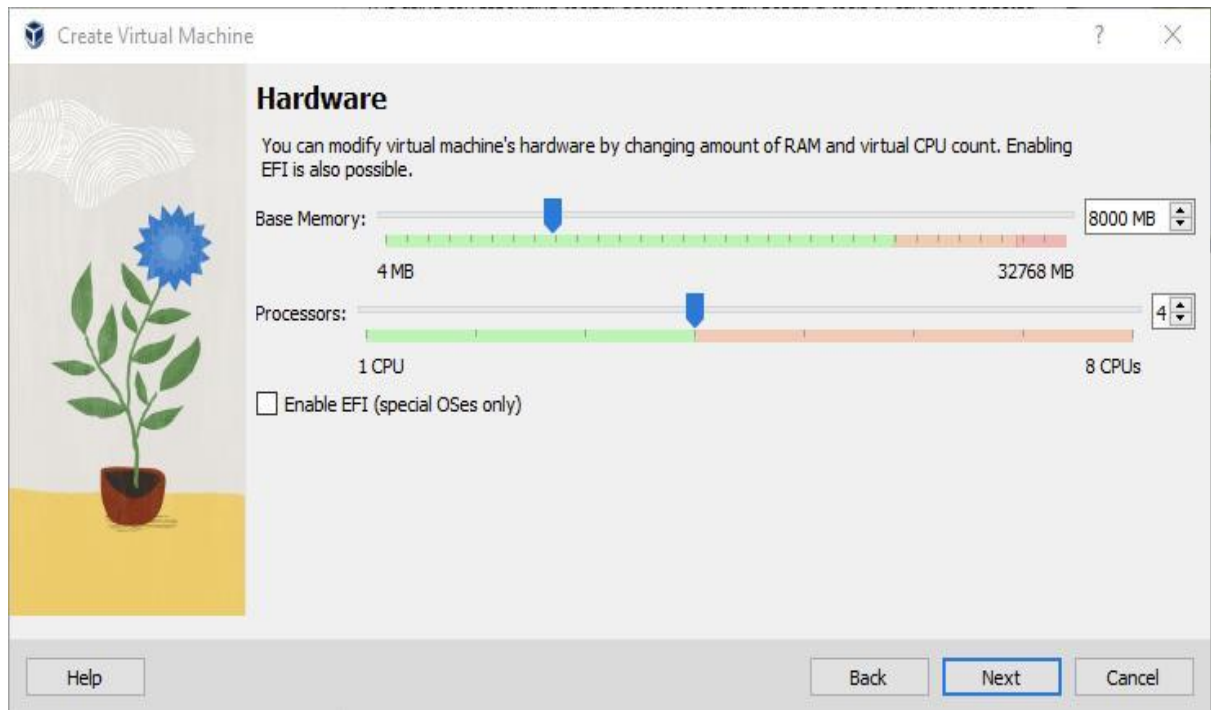
AIM:

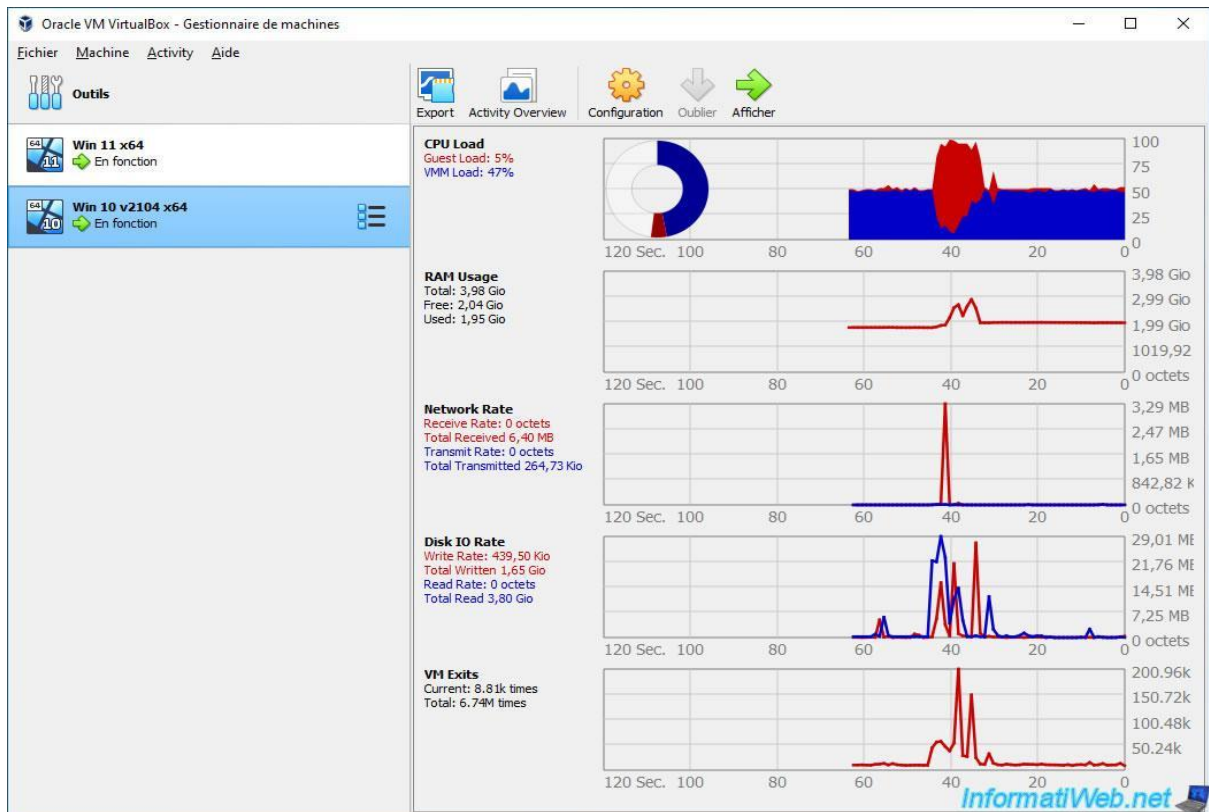
To study the procedure for running virtual machines with different hardware configurations and determine how many virtual machines can be utilized simultaneously on a host computer based on available RAM, CPU and storage resources.

PRODECURE:

1. Open **Oracle VirtualBox** on the host computer.
2. Create the first virtual machine by clicking **New** and select the required operating system.
3. Allocate a suitable amount of **RAM, CPU and virtual hard-disk space** for the first VM.
4. Create and start the first virtual machine and verify that the guest OS is running.
5. Create a second virtual machine with a **different configuration**, such as different RAM, CPU and storage.
6. Similarly, create additional virtual machines with different configurations.
7. Start the virtual machines one by one and observe their performance.
8. Run multiple virtual machines **simultaneously** and monitor the available host RAM and CPU resources.
9. Open **VirtualBox Manager** → **Resources/Resource Use** to observe the resource usage of the running virtual machines. VirtualBox provides resource metrics for individual VMs and the host system.
10. Continue starting additional VMs until the host system reaches its practical resource limit or a VM cannot start because sufficient RAM is unavailable.
11. Record the **number of virtual machines running simultaneously**, along with their RAM and CPU configurations.
12. Compare the configurations and determine how the number of simultaneously usable VMs depends on the available **RAM, CPU and storage**. VirtualBox documentation specifically warns that insufficient RAM can prevent another VM from starting.

OUTPUT:





Tools

New Settings Discard Show

ovs34-efi Powered Off

ubuntu-18.04 Running

win2016srv Running

centos7 Powered Off

OracleLinux7 Running

OracleLinux6 Powered Off

ol7-vbox6 Powered Off

General
Name: ubuntu-18.04
Operating System: Ubuntu (64-bit)
Settings File Location: /Users/scr/VirtualBox/ubuntu-18.04

System
Base Memory: 4096 MB
Boot Order: Optical, Hard Disk
Acceleration: VT-x/AMD-V, Nested Paging, KVM Paravirtualization

General
Name: win2016srv
Operating System: Windows 2016 (64-bit)
Settings File Location: /Users/scr/VirtualBox/win2016srv

System
Base Memory: 8192 MB
Boot Order: Hard Disk, Optical, Floppy
Acceleration: VT-x/AMD-V, Nested Paging, PAE/NX

Preview

Preview